

05.1	1. Use a grid OR Divide area into squares/sections/plots; 2. Method of obtaining random coordinates/numbers; 3. Use same method of measurement (of the plants each time); 4. Repeat a large number of times OR Measure a large number of plants; 5. (Find the) sum of heights and divide by number of plants; 6. Calculate a running mean until number becomes (fairly) constant;	5 max (5 x AO1)	1. Accept use tape measures for 'use a grid' 2. Accept calculator/computer/random numbers table/generator for method 3. Accept descriptions of measurements eg from soil/ground to highest leaf or to top of plant or to (shoot) tip or to highest part/bit 4. Accept ≥ 10 for large number 4. Accept many/multiple but ignore several 5. Accept total for sum 5. Accept measurements for heights or plants 5. Accept a word equation eg <u>Height total</u> Number of plants
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Question	Marking Guidance	Mark	Comments
05.2	1. Plant height and /cm in first column and Frequency (density), in second column; 2. Non-overlapping class intervals between 60 and 120; eg $60 \leq x < 70$ OR 60 – 69, 70 – 79 etc OR 60 – 69.9, 70 – 79.9 etc OR 59.5 – 69.0, 69.5 – 79 etc	2 (2 x AO2)	1. Accept <i>M. sylvestris</i> for plant 1. Accept length for height 1. Accept mm or m for cm 1. Accept number for frequency 2. Accept unequal class widths if frequency density used in heading 2. Accept correct figures in either column 2. Figures given must match units in heading, eg 600 if heading shows / mm or 0.6 if heading shows /m 2. Accept fewer or more than 6 rows of data if data covers a range between 60 and 120 2. Ignore any numbers in the frequency column

Question	Marking Guidance	Mark	Comments
05.3	Correct answer of 2.11 = 2 marks;; Incorrect answer that shows correct total in numerator 1561250 = 1 mark OR Correct total in denominator 741250 = 1 mark;	2 (2 x AO1)	For 2 marks, accept 2.1(0623946) or any correct rounding eg 2.1 or 2.106 etc

Question	Marking Guidance	Mark	Comments
05.4	(In relation to wild flower fields) 1. More plant species OR Larger plant (index of bio)diversity; 2. (So) more variety of food/nectar/pollen OR (So) more (types of) habitat/niche OR (So) more/better protection from predators;	2 (2 x AO2)	Accept the converse in the context of wheat fields 1. Accept more types/variety of plants 1. Accept greater species richness of plants 2. Ignore more/better food but accept more food sources 2. Ignore more shelter 2. Ignore more homes 2. Accept 'types of' for 'variety of' 2. Accept 'fewer predators' for 'more protection'